BUSINESS MANAGEMENT SYSTEM (BMS)

Course: INSY281_System Analysis & Design

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1. Executive Summary

This Baseline Project Plan proposes the development of a Business Management System (BMS) for a medium-sized trading company. The company currently uses paper files and separate spreadsheets to manage sales, purchases, inventory, suppliers, customers and financial summaries. This fragmented process can lead to duplicate records, slow reporting, stock discrepancies and delayed decisions. The proposed BMS will create a centralized system for recording transactions and providing timely management reports.

The project is expected to take 16 weeks with an estimated budget of KSh 1,850,000. An iterative Agile development approach will be used, allowing stakeholders to review working features in short cycles. The system type is a transaction-processing and management information system with role-based access. The deliverables are a requirements specification, database and application, tested modules, user documentation, training, deployment and a GitHub repository containing the project report and approved source code.

2. Background, Problem Statement and Opportunity

The fictional company, Horizon Traders Ltd., sells household and office products through a small warehouse and sales office. Staff maintain separate records for stock, orders, customer accounts and supplier purchases. Managers need to combine these records manually to understand daily sales, stock levels and operating performance. Manual processes make it hard to track changes, identify low-stock items and produce consistent reports.

Problem statement: Disconnected records and manual processing increase the risk of inaccurate data, stock-outs, duplicated work and delayed business decisions.

Opportunity: A centralized BMS can help standardize business processes, give visibility to stock and transactions and ensure reliable operational reports are available to authorized staff.

3. Project Goal, Objectives and Scope

Goal: Deliver a secure, usable system that integrates core daily operations for Horizon Traders Ltd.

Measurable objectives:

- Reduce the time required to prepare a daily sales and stock summary to under 5 minutes.
- Record at least 95% of sales, purchases and stock movements in the system during the first month after rollout.
- Give alerts for low stock for configured reorder levels.
- Restrict sensitive functions according to user roles and maintain an audit trail for key transactions.
- Train all designated users before deployment and achieve at least 80% on a post-training task checklist.

In scope: user authentication and roles; product and inventory records; supplier and customer records; sales and purchase transactions; stock movement history; basic expense recording; dashboards and standard reports; backup and restore procedures.

Out of scope: full payroll, advanced tax filing, multi-country operations, direct bank integration, mobile native apps and automated delivery logistics. These may be considered in a later phase.

4. System Type and Development Approach

The proposed solution is a Transaction Processing System (TPS) for daily sales, purchasing and inventory transactions plus a Management Information System (MIS) for summaries and reports. It will be a web-based application using a centralized relational database and authorize staff access to features based on assigned roles, such as administrator, manager, sales clerk and storekeeper.

The project team will employ an iterative Agile approach. Work will be split into short development iterations, each producing demonstrable features. The team will seek feedback from the owner and staff, prioritize changes, test each increment and update the backlog. This approach is suitable because users can clarify workflow requirements after trying prototypes. Formal approval will still be required at scope, testing and deployment milestones.

5. Stakeholders and Functional Requirements

Key stakeholders include the company owner/sponsor, project manager or systems analyst, developer(s), sales staff, storekeeper, accountant/finance representative and system administrator.

The sponsor approves scope and funding; users validate the workflows; the technical team designs, builds, tests and documents the system.

Core functional requirements:

1. Users sign in securely and receive permissions based on role.
2. Authorized staff create, view, update and deactivate product, customer and supplier records.
3. The system records sales and purchase transactions with dates, quantities, prices and responsible users.
4. Confirmed sales reduce stock and received purchases increase stock; corrections must be traceable.
5. The system flags products whose available quantity reaches the configured reorder level.
6. Managers can generate daily, weekly and monthly sales, purchase, inventory and expense reports.
7. The system records audit events for critical actions and supports scheduled backups.

Non-functional requirements: common pages load within 3 seconds under normal test conditions; the system supports at least 25 concurrent users in acceptance testing; data is backed up daily; passwords are stored using a secure one-way hashing method; and the interface is usable on standard desktop and mobile browsers. These targets will be verified during testing.

6. Feasibility Analysis

Schedule feasibility: A 16-week plan is achievable if the sponsor provides timely decisions, users attend reviews and the scope remains controlled. Delayed approvals or major scope changes may require re-planning.

Legal and contractual feasibility: The company must follow applicable Kenyan data-protection and business-record requirements, limit access to personal and commercial data and agree on software licensing, hosting terms and vendor responsibilities. Specific compliance obligations should be confirmed with a qualified adviser before deployment.

Political and organizational feasibility: Staff may resist changes to familiar manual processes or worry about monitoring. Involving users early, explaining benefits, documenting responsibilities and offering training will reduce resistance.

7. Deliverables and Acceptance Criteria

The project will have the following deliverables:

1. Project initiation and BPP

The project scope, objectives, assumptions, budget, schedule, and risks must be defined and approved by the project sponsor.

2. Requirements and process models

Functional and non-functional requirements must be reviewed and approved by representative users.

3. System design

The database schema, system architecture, screen prototypes, and user access roles must be recorded.

4. Working Business Management System (BMS) application

The application modules must be working as per the test cases and all critical defects resolved.

5. Test evidence

The evidence includes unit, integration, security, and user testing documentation and no critical defects at release.

6. User guide and training

The user guide should be available and the designated users complete the training checklist.

7. Deployment and handover

The configuration of the production environment, backup and restoration procedures, and the project sponsor approval for handover.

8. GitHub repository and final report

The GitHub link should be included in the final report. The final report should be submitted in PDF format and together with all the permitted project files.

8. Project Schedule and Milestones

The project schedule assumes that the work will start only after sponsor approval. The weeks should be calculated from the approved project start date.

1. Initiation and stakeholder interviews (Weeks 1–2)

This stage will see the project team initiating the project, carrying out stakeholder interviews, and developing the Business Project Proposal (BPP) and initial risk register. The final evidence of this stage is the approved BPP and risk register.

2. Requirements analysis and feasibility (Weeks 3–4)

This stage will entail gathering and analyzing the requirements and reviewing the feasibility of the project. The evidence for this stage completion includes signed requirements and a feasibility review.

3. System and database design (Weeks 5–6)

The team will carry out the system and database design, including application prototyping. This process must have approvals for the designs and the prototype.

4. Iteration 1: Users, products and suppliers (Weeks 7–8)

This stage will focus on completing the first development iteration to include user management, product management, and supplier management. The team will present the modules and record user feedback.

5. Iteration 2: Sales, purchases and inventory (Weeks 9–11)

The second iteration will be about completing the second set of modules in the application, including sales, purchases, and inventory. Completion of this stage will be evidenced by the demonstrated integrated transaction modules.

6. Iteration 3: Reports, audit and backups (Weeks 12–13)

This iteration will be about completing the reporting, audit records, and backups in the application. The team will demonstrate the reports and backup procedures during this stage.

7. Testing, fixes and user acceptance (Week 14)

This stage will involve testing the application, fixing the identified defects, and obtaining formal user acceptance. The evidence for this stage will be a test report and user acceptance approval.

8. Training, deployment and handover (Week 15)

The users will be trained, and the system deployed. A training checklist completed by users and a completed deployment checklist will serve as evidence of this stage's completion.

9. Final review and closure (Week 16)

This stage entails carrying out the project team's final review, preparing the final report, submitting the GitHub repository, and completing the lessons learned records.

9. Estimated Project Budget

The company estimates a total project budget amounting to KSh 1,850,000 for the Business Management System.

This budget breakdown is as follows:

1. Requirements analysis and project management: KSh 250,000
2. System design and database design: KSh 180,000
3. Application development: KSh 620,000
4. Testing and quality assurance: KSh 180,000
5. Hosting, domain, tools, and security setup: KSh 120,000
6. Data preparation and migration: KSh 100,000
7. Training and documentation: KSh 90,000
8. Contingency reserve: KSh 310,000

Total: KSh 1,850,000.

All the costs must be validated and approved before spending. Any costs incurred from the contingency reserve must be approved by the project sponsor.

10. Risks and Mitigation

There are several risks that may affect this project. The following strategies will minimize them.

1. Requirements changes

This risk entails changes in the initial requirements by the users. To control this risk, the team will use a prioritized backlog of requirements and have every change requested in writing.

2. Inaccurate legacy data

The data may be inaccurate or have duplicates in the existing systems. We will mitigate this risk by profiling the data, removing duplicates, and testing sample data imports with users.

3. User resistance

The project users may be reluctant to adopt the new system. We will control this risk by engaging the staff during demonstrations and ensuring they understand their roles during training.

4. Data loss or unauthorized access

Data loss or theft might occur, especially if the wrong people access the system. The system will have access credentials for different users with strong passwords, backup procedures, and restoration testing to control this risk.

5. Schedule delays

Some activities may take longer than anticipated and cause delays. We will mitigate this risk by frequently reviewing schedule milestones and addressing problems as early as possible.

6. Budget overrun

Some project costs may be more than the initial budget and cause a deficit. We will control this risk by monitoring expenditures against the approved budget and having scope changes reviewed and authorized.

11. Communication, Monitoring and Control

The project team will constantly communicate with the project sponsor, stakeholders, developers, and users throughout the project's duration.

Communication: The project team will hold regular project-related meetings and report on project progress to the relevant stakeholders. All communication must go through the project manager.

Monitoring: The project deliverables, progress, schedule, budget, and scope will be monitored to ensure they meet the quality requirements. This will be done by frequently reviewing the project's completion against the approved schedule, budget, scope, and quality benchmarks.

Control: There must be a formal change control process that requires any significant changes to the project's scope, schedule, or budget to be reviewed and approved by the project sponsor before implementation can take place.

Reporting: The project manager will provide progress reports to the sponsor and stakeholders. These reports should indicate the completed and pending activities, risks, and corrective measures.

Communication, monitoring, and control will ensure the Business Management System is completed within the approved budget and time and meets the users' needs.

12. Conclusion

The proposed Business Management System is intended to improve the consistency and visibility of daily sales, purchasing and inventory operations at Horizon Traders Ltd. The 16-week iterative plan, defined acceptance criteria, feasibility review and risk controls, and the KSh 1,850,000 estimated budget provide a baseline for monitoring implementation. The project should only be taken up by the sponsor once they have validated the requirements, confirmed the funding and approved the schedule.